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Image-to-image translation between fracture types in femur X-rays using generative adversarial networks and diffusion models

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Abstract

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1 Introduction

1.1 Motivation

Atypical femoral fractures (AFFs) are a rare type of fractures in the femur. Unlike normal femoral fractures (NFFs), they are not caused by high immediate stress, but are related to long-term bisphosphonate medication prescribed to patients with osteoporosis [1]. The identification of AFFs remains challenging in clinical practice, and these fractures are frequently overlooked unless explicitly assessed. Less than 7% of all cases are correctly diagnosed in radiology reports [2]. In precision medicine and personalized healthcare, accurate diagnosis is crucial for effective treatment. AFFs have a high rate of reopertion [3] and therefore require a special focus. To avoid unecessary surgeries, reliable and early detection is essential.

To support medical doctors in improving diagnostic accuracy, deep neural networks have been trained to classify AFFs and NFFs [2]. However, the dataset used in previous studies was relatively small, consisting of approximately 1,000 images from around 375 patients. In this thesis, we utilize a larger dataset comprising approximately 4300 images from around 1200 patients. Nevertheless, this dataset presents a common challenge in medical imaging: severe class imbalance. The minority class (AFF) is significantly underrepresented, a well known issue in machine learning, that can negatively affect model performance [4]. Several strategies have been proposed to address class imbalance. Oversampling increases the number of minority class samples, often by duplicating existing images, while undersampling reduces the number of majority class samples. Although these techniques can improve performance, they also introduce potential drawbacks, such as overfitting in the case of oversampling and information loss in the case of undersampling.

With the rapid development of generative models, alternative approaches for dataset augmentation have emerged. Instead of reusing existing images, synthetic samples can be generated by learning the underlying high-dimensional data distribution.

Over the past decade, generative adversarial networks (GANs) have been the dominant architecture for generative tasks in medical imaging [5]. More recently, different versions of diffusion models showed remarkable preformance in various areas and have increasingly been applied to medical imaging tasks [6].

Generative models can be conditioned in different ways during the synthesis process. Previous thesis work used noise as input to generate synthetic images [7] in so-called noise-to-image processing. This thesis focuses on image-to-image translation, where the model

learns to translate images from one domain to another. This allows a more controlled synthesis conditioned on existing anatomical structures. A well established architecture for this task is CycleGAN [8]. We compare this approach with a diffusion-based model optimized for image-to-image translation, CycleDiff [9].

1.2 Research questions

The aim of this thesis is to establish whether synthesising data using generative models can improve the classification of rare fracture types. For this purpose, a CycleGAN and a CycleDiff model are trained on X-rays of AFFs and NFFs to create balanced datasets. The generative models are evaluated based on how well the generated images imitate the real data.

To assess whether generative augmentation improves fracture classification, a ResNet-50 classifier is trained under different conditions. First, a baseline model is trained using only the real data. Second, a classifier is trained using real data augmented with synthetically generated images. The performance of these models is then compared using standard metrics like accuracy and the F1-score and, additionally, using statistical tests.

Furthermore, to determine whether generative augmentation provides an advantage over conventional data augmentation techniques (such as rotation, flipping, and cropping), a ResNet-50 trained on real data augmented with standard methods is compared to a ResNet-50 trained on real data augmented with generative models.

1. Does CycleGAN or CycleDiff perform better in the task of generating synthetic images by transforming between AFF and NFF images?
2. Does a ResNet-50 classifier trained on real data augmented with generated images perform significantly better than a classifier trained only on real data?
3. Does generative augmentation lead to significantly better classification performance compared to the standard augmentation techniques rotation, flipping and cropping?

1.3 Delimitations

tbd

This is where the main delimitations are described. For example, this could be that one has focused the study on a specific application domain or target user group. In the normal case, the delimitations need not be justified.

2 Theory

Applications of AI led to improvements in different areas in healthcare . The quality of these solutions depends on the data that they are trained on. For most models it has to have a high volume and quality. Due to different issues, such datasets are seldom available: privacy issues and limited sharing between hospitals and research institutions make it hard to gather medical data in the first place. Then, annotating the data is expensive, as trained medical personal is required to assure a high quality of labels [10].

enter examples

An approach to create bigger and high quality datasets is to use generative models to create synthetic data that is highly similar to real data.

include related work that showed this

This chapter explains the foundations of generative models and different architectures. It also covers the theory behind image classification models that are used to evaluate the capability to use the synthetic data to augment real data for a downstream task.

2.1 Generative Models

The general goal of generative models is to approximate an unknown distribution $p_{data}(\boldsymbol{x})$ of the training data $\boldsymbol{x}$ with a learned distribution $p_{model}(\boldsymbol{x})$ [11]. p_{model} can then be used to draw new samples, that could also come from p_{data}. There are different ways to train a model to learn this distribution. The most popular approaches are Generative Adversarial Networks (GANs) and Diffusion Models (DMs).

Different types of input can be given to the models to create a synthetic image. The basic case is to use noise as a starting point. The noise vector $\boldsymbol{z}$ follows a distribution $p(\boldsymbol{z})$, usually a high-dimensional Gaussian. These models are called noise-to-image models. Image-to-image models take, as the name indicates, another image as input. It is used to translate from a source domain to an output domain [12] and finds application in e.g. image inpainting, image colorization or to increase the resolution of an image. Popular are also text-to-image models that can create images from a text prompt.

In the case of **image-to-image translation**, we assume two unknown data distributions $p_{data}(\boldsymbol{x})$ and $p_{data}(\boldsymbol{y})$. The model needs to learn a function on how to translate between domain X of the one distribution and domain Y of the other distribution (see formula 2.1).

$$F : X \rightarrow Y \text{ such that } F(x) \sim p_{data}(y) \tag{2.1}$$

Ideally, the data consists of paired images, i.e. every image in domain X has a corresponding image in domain Y. In this case, the network can directly learn from the joined probability distribution. For unpaired images, the model needs to learn from the two marginal distributions, which is more difficult.

In the following, the two most popular types of architectures are explained on a basic level and how they can be used for image-to-image translation.

2.1.1 Generative Adversarial Networks

For a long time , GANs dominated the field of synthetic image generation. They were introduced by Goodfellow et al. [11] in 2014 and are the foundation for many adjustments and improvements until today.

sounds weird, but also kind of true

The basic idea is to have two adversarial networks that try to outperform each other. On the one side stands the generator $G(z; \Theta^G)$. z is the input noise vector and the generator tries to learn how transform it into a realistic image. Θ^G is a set of parameters that are learned by the generator to improve its ability to fool its counter part, which is the discriminator $D(z; \Theta^G)$. Θ^D. The discriminator tries to classify input correctly as real or synthetic images. It also has a set of parameters Θ^D that are learned during the training process.

An adversarial loss function is defined (see formula 2.2), that the generator tries to minimize and the discriminator tries to maximize.

$$\min_G \max_D [\mathbb{E}_{x \sim p_{\text{data}}(x)}[\log D(x)] + \mathbb{E}_{z \sim p_z(z)}[\log(1 - D(G(z)))]] \tag{2.2}$$

In this basic case, a synthetic image is created from the noise vector z but for image-to-image translation one image is transformed into an adjusted one in the target domain.

2.1.2 Diffusion Models

from first idea to CycleDiff

2.1.3 Metrics

FID/IS

Radiometrics + statistical test

2.2 Image Classification

few general words about history of image classification, in general and medical context

2.2.1 Convolutional Neural Networks

2.2.2 Metrics

3 Method

In this chapter, the method is described in a way which shows how the work was actually carried out. The description must be precise and well thought through. Consider the scientific term replicability. Replicability means that someone reading a scientific report should be able to follow the method description and then carry out the same study and check whether the results obtained are similar. Achieving replicability is not always relevant, but precision and clarity is.

Sometimes the work is separated into different parts, e.g. pre-study, implementation and evaluation. In such cases it is recommended that the method chapter is structured accordingly with suitable named sub-headings.

4 Results

This chapter presents the results. Note that the results are presented factually, striving for objectivity as far as possible. The results shall not be analyzed, discussed or evaluated. This is left for the discussion chapter.

In case the method chapter has been divided into subheadings such as pre-study, implementation and evaluation, the result chapter should have the same sub-headings. This gives a clear structure and makes the chapter easier to write.

In case results are presented from a process (e.g. an implementation process), the main decisions made during the process must be clearly presented and justified. Normally, alternative attempts, etc, have already been described in the theory chapter, making it possible to refer to it as part of the justification.

5 Discussion

This chapter contains the following sub-headings.

5.1 Results

Are there anything in the results that stand out and need be analyzed and commented on? How do the results relate to the material covered in the theory chapter? What does the theory imply about the meaning of the results? For example, what does it mean that a certain system got a certain numeric value in a usability evaluation; how good or bad is it? Is there something in the results that is unexpected based on the literature review, or is everything as one would theoretically expect?

5.2 Method

This is where the applied method is discussed and criticized. Taking a self-critical stance to the method used is an important part of the scientific approach.

A study is rarely perfect. There are almost always things one could have done differently if the study could be repeated or with extra resources. Go through the most important limitations with your method and discuss potential consequences for the results. Connect back to the method theory presented in the theory chapter. Refer explicitly to relevant sources.

The discussion shall also demonstrate an awareness of methodological concepts such as replicability, reliability, and validity. The concept of replicability has already been discussed in the Method chapter (3). Reliability is a term for whether one can expect to get the same results if a study is repeated with the same method. A study with a high degree of reliability has a large probability of leading to similar results if repeated. The concept of validity is, somewhat simplified, concerned with whether a performed measurement actually measures what one thinks is being measured. A study with a high degree of validity thus has a high level of credibility. A discussion of these concepts must be transferred to the actual context of the study.

The method discussion shall also contain a paragraph of source criticism. This is where the authors' point of view on the use and selection of sources is described.

In certain contexts it may be the case that the most relevant information for the study is not to be found in scientific literature but rather with individual software developers and open

source projects. It must then be clearly stated that efforts have been made to gain access to this information, e.g. by direct communication with developers and/or through discussion forums, etc. Efforts must also be made to indicate the lack of relevant research literature. The precise manner of such investigations must be clearly specified in a method section. The paragraph on source criticism must critically discuss these approaches.

Usually however, there are always relevant related research. If not about the actual research questions, there is certainly important information about the domain under study.

5.3 The work in a wider context

There must be a section discussing ethical and societal aspects related to the work. This is important for the authors to demonstrate a professional maturity and also for achieving the education goals. If the work, for some reason, completely lacks a connection to ethical or societal aspects this must be explicitly stated and justified in the section Delimitations in the introduction chapter.

In the discussion chapter, one must explicitly refer to sources relevant to the discussion.

6 Conclusion

This chapter contains a summarization of the purpose and the research questions. To what extent has the aim been achieved, and what are the answers to the research questions?

The consequences for the target audience (and possibly for researchers and practitioners) must also be described. There should be a section on future work where ideas for continued work are described. If the conclusion chapter contains such a section, the ideas described therein must be concrete and well thought through.

Bibliography

[1] Jörg Schilcher, Karl Michaëlsson, and Per Aspenberg. "Bisphosphonate Use and Atypical Fractures of the Femoral Shaft". In: *New England Journal of Medicine* 364.18 (May 2011). _eprint: https://www.nejm.org/doi/pdf/10.1056/NEJMoa1010650, pp. 1728–1737. ISSN: 0028-4793. DOI: 10.1056/NEJMoa1010650. URL: https://www.nejm.org/doi/full/10.1056/NEJMoa1010650 (visited on 02/14/2026).

[2] Georg Zdolsek, Yupei Chen, Hans-Peter Bögl, Chunliang Wang, Mischa Woisetschläger, and Jörg Schilcher. "Deep neural networks with promising diagnostic accuracy for the classification of atypical femoral fractures". In: *Acta Orthopaedica* 92.4 (July 2021). _eprint: https://doi.org/10.1080/17453674.2021.1891512, pp. 394–400. ISSN: 1745-3674. DOI: 10.1080/17453674.2021.1891512. URL: https://doi.org/10.1080/17453674.2021.1891512 (visited on 01/20/2026).

[3] H.P. Bögl, K. Michaëlsson, G. Zdolsek, J. Höijer, and J. Schilcher. "Increased rate of reoperation in atypical femoral fractures is related to patient characteristics and not fracture type. A nationwide cohort study". en. In: *Osteoporosis International* 31.5 (May 2020), pp. 951–959. ISSN: 1433-2965. DOI: 10.1007/s00198-019-05249-3. URL: https://doi.org/10.1007/s00198-019-05249-3 (visited on 02/16/2026).

[4] Arman Ghavidel and Pilar Pazos. "Machine learning (ML) techniques to predict breast cancer in imbalanced datasets: a systematic review". en. In: *Journal of Cancer Survivorship* 19.1 (Feb. 2025), pp. 270–294. ISSN: 1932-2267. DOI: 10.1007/s11764-023-01465-3. URL: https://doi.org/10.1007/s11764-023-01465-3 (visited on 02/16/2026).

[5] Mohd Ali, Mehboob Ali, Mubashir Hussain, and Deepika Koundal. "Generative Adversarial Networks (GANs) for Medical Image Processing: Recent Advancements". en. In: *Archives of Computational Methods in Engineering* 32.2 (Mar. 2025), pp. 1185–1198. ISSN: 1886-1784. DOI: 10.1007/s11831-024-10174-8. URL: https://doi.org/10.1007/s11831-024-10174-8 (visited on 01/21/2026).

[6] Yaqing Shi, Abudukelimu Abulizi, Hao Wang, Ke Feng, Nihemaiti Abudukelimu, Youli Su, and Halidanmu Abudukelimu. "Diffusion Models for Medical Image Computing: A Survey". In: *Tsinghua Science and Technology* 30.1 (Feb. 2025), pp. 357–383. ISSN: 1007-0214. DOI: 10.26599/TST.2024.9010047. URL: https://ieeexplore.ieee.org/abstract/document/10676408 (visited on 01/20/2026).

[7] Yaning Wang. *Synthetic X-ray Images of Femur Fractures*. eng. 2025. URL: https://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-218223 (visited on 01/21/2026).

[8] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros. *Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks*. arXiv:1703.10593 [cs] version: 6. Nov. 2018. DOI: 10.48550/arXiv.1703.10593. URL: http://arxiv.org/abs/1703.10593 (visited on 01/20/2026).

[9] Shilong Zou, Yuhang Huang, Renjiao Yi, Chenyang Zhu, and Kai Xu. *CycleDiff: Cycle Diffusion Models for Unpaired Image-to-image Translation*. en. arXiv:2508.06625 [cs]. Jan. 2026. DOI: 10.48550/arXiv.2508.06625. URL: http://arxiv.org/abs/2508.06625 (visited on 02/11/2026).

[10] Mahmoud Ibrahim, Yasmina Al Khalil, Sina Amirrajab, Chang Sun, Marcel Breeuwer, Josien Pluim, Bart Elen, Gökhan Ertaylan, and Michel Dumontier. "Generative AI for synthetic data across multiple medical modalities: A systematic review of recent developments and challenges". In: *Computers in Biology and Medicine* 189 (May 2025), p. 109834. ISSN: 0010-4825. DOI: 10.1016/j.compbiomed.2025.109834. URL: https://www.sciencedirect.com/science/article/pii/S0010482525001842 (visited on 01/21/2026).

[11] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. "Generative adversarial networks". In: *Commun. ACM* 63.11 (Oct. 2020), pp. 139–144. ISSN: 0001-0782. DOI: 10.1145/3422622. URL: https://dl.acm.org/doi/10.1145/3422622 (visited on 01/28/2026).

[12] Yingxue Pang, Jianxin Lin, Tao Qin, and Zhibo Chen. "Image-to-Image Translation: Methods and Applications". In: *IEEE Transactions on Multimedia* 24 (2022), pp. 3859–3881. ISSN: 1941-0077. DOI: 10.1109/TMM.2021.3109419. URL: https://ieeexplore.ieee.org/abstract/document/9528943 (visited on 03/05/2026).